

ANN MARY THOMAS

Energy Analyst · Measurement & Verification · Deep Learning Researcher

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★ **PUBLISHED** Nature npj Urban Sustainability (July 2026) · *TAL-Net: a temporal attention LSTM framework for urban electricity and emissions forecasting in the U.S.* - DOI: 10.1038/s42949-026-00424-y

PROFESSIONAL SUMMARY

Energy Analyst with expertise in Measurement & Verification (M&V), data-driven energy performance modelling, and deep learning applied to power grid systems. Published first author in Nature npj Urban Sustainability. Track record spanning clean energy analytics at Ameresco (NYSE: AMRC), a £14.1M Innovate UK programme at WMG / University of Warwick, and marine emissions modelling at Navalt. Proficient in Python, Power BI, IPMVP, and ISO 50001 - consistently turning complex energy datasets into measurable sustainability outcomes.

PROFESSIONAL EXPERIENCE

Energy Analyst

Ameresco, Inc. (NYSE: AMRC)

Measurement & Verification (M&V) Team · UK & Europe

Jun 2026 – Present

London, UK

- ▶ **Develop** and maintain **data-driven energy performance models** using utility, metering, and operational datasets to track and evidence energy and carbon savings across client projects
- ▶ **Analyse** large, multi-source datasets to identify consumption trends, anomalies, and optimisation opportunities; contribute M&V plans aligned to **IPMVP methodology**
- ▶ **Build automated Python data pipelines** to improve delivery efficiency; produce Power BI dashboards and technical reports for technical and non-technical stakeholders

Research Assistant - Energy Forecasting & Carbon Analytics

GENESIS Research Lab, London Metropolitan University

Under Dr. Maitreyee Dey

Feb 2025 – Present (pt)

London, UK

2 papers co-authored **0.48%** MAPE demand **1.60%** MAPE CO₂ **19×** vs ARIMA **127k** car-equiv. CO₂ cut

- ▶ **Independently** designed **TAL-Net** - hybrid bidirectional LSTM + temporal attention model - on US EIA-930 grid data (California & Texas); achieved **0.48% MAPE (demand)** and **1.60% MAPE (CO₂)**, a **19× improvement over ARIMA**; first paper published in Nature npj Urban Sustainability (July 2026)
- ▶ **Engineered** 27 input features; SHAP confirmed coal (68.38%) and natural gas (23.99%) as dominant CO₂ drivers; **50% solar penetration scenario yielded 6.74% CO₂ reduction** (~127k cars); all results p < 0.0001

Energy Consumption & Sustainability Analyst

WMG, University of Warwick - Project STRATA

Dec 2025 – Apr 2026

Coventry, UK

Project STRATA - £14.1M ATI / Innovate UK · Honeywell Consortium (National Centre for Connected Manufacturing). Deployed at 3T Additive Manufacturing.

£14.1M programme **28,510 kWh** baseline **6.65 tCO₂** carbon **£9,907** annual cost **15–18%** projected saving **35+** equipment types

- ▶ Built complete energy performance baseline for 35-staff manufacturing facility across 35+ equipment types from zero (no sub-meters/BMS); IPMVP Option A; validated against ND-NEED 2022, CIBSE Guide F; identified **HVAC (49%), boiler (25%), lighting (12%)** as primary hotspots
- ▶ Designed **6 decarbonisation scenarios** with full ROI and 5-year financial projections (combined **15–18% annual energy reduction**); policy framework aligned to **ISO 50001, ISO 14001, ISO 14040/44**
- ▶ Built Power BI dashboards (kWh/employee, kWh/m², carbon intensity); presented findings to WMG management, 3T leadership, and NCC/Honeywell consortium partners

Energy Consumption & Sustainability Analyst (Intern)

3T Additive Manufacturing - WMG Industry Placement

Sep 2025 – Dec 2025

Coventry, UK

39.3% carbon reduction **6** scenarios **16%** shutdown saving **59%** kitchen sub-category **5-year** ROI projections

- ▶ **Achieved 39.3% carbon footprint reduction** through systematic energy baseline analysis, load factor modelling (IPMVP Option A), and 6 targeted improvement scenarios with 15–18% combined annual reduction projection
- ▶ Internship led directly to analyst role at WMG, University of Warwick, continuing STRATA programme support

Junior Data Analyst - Energy & Emissions Performance

Navalt Solar and Electric Boats Pvt. Ltd

Aug 2022 – Jan 2024

Remote

700+ vessels **200,000+** dataset rows **85%** efficiency gain **1wk→4hrs** reporting **4 docs** → CII Guide

- ▶ Built **automated Python ETL pipeline** processing 700+ vessels (200,000+ rows, 200+ metrics) - reducing quarterly ESI reporting from ~1 week to **~4 hours (85% efficiency gain)**
- ▶ Performed **CII (IMO)** calculations for the full fleet; calculated SFOC/SFOS metrics; submitted quarterly ESI compliance reports to port authorities; condensed 4 IMO regulatory documents into a single CII Reference Guide

PUBLICATION

TAL-Net: a temporal attention LSTM framework for urban electricity and emissions forecasting in the U.S.

Ann Mary Thomas, Maitreyee Dey & Soumya Prakash Rana · GENESIS Lab, London Metropolitan University

npj Urban Sustainability (Nature Publishing Group) · Published 17 July 2026 · Open Access · DOI: 10.1038/s42949-026-00424-y

SKILLS

ENERGY & SUSTAINABILITY

Measurement & Verification · IPMVP · Energy Baselines · Load Factor Modelling · ISO 50001 · ISO 14001 · CIBSE Guide F · CII (IMO) · Carbon Accounting

AI / ML / DEEP LEARNING

TensorFlow / Keras · LSTM & Attention · Bidirectional LSTM · RBM + MLP · scikit-learn · SHAP · Time-Series Forecasting · Feature Engineering

LANGUAGES & DATA

Python · SQL · pandas · NumPy · Matplotlib · Seaborn · Jupyter · Git / GitHub

PLATFORMS & REPORTING

Power BI · Excel (Advanced) · VS Code · Statistical Testing · Technical Report Writing · Data Storytelling

EDUCATION

MSc Data Analytics

London Metropolitan University

2024 – 2025

London, UK

Deep learning research at GENESIS Lab; designed TAL-Net; co-authored 2 peer-reviewed papers. First paper published in Nature npj Urban Sustainability (July 2026).

B.Tech in Mechanical Automobile Engineering

SCT College of Engineering

2017 – 2021

Foundation in mechanical systems, thermodynamics, and engineering fundamentals.

CERTIFICATIONS

- ▶ Excel for Data Analytics · Simplilearn
- ▶ Statistics for Data Science · Simplilearn

KEY ACHIEVEMENTS

- ▶ **Published in Nature npj Urban Sustainability (July 2026)** - First author on TAL-Net; DOI: 10.1038/s42949-026-00424-y
- ▶ **39.3% carbon footprint reduction** at 3T Additive Manufacturing through systematic data-driven energy analysis
- ▶ **19× accuracy improvement over ARIMA** - TAL-Net achieves 0.48% MAPE on EIA-930 electricity demand forecasting
- ▶ Delivered **28,510 kWh / 6.65 tCO₂ / £9,907 facility baseline** for £14.1M Innovate UK STRATA, validated against ND-NEED 2022 and CIBSE Guide F
- ▶ **85% efficiency gain** in ESI regulatory reporting at Navalt through Python ETL automation (1 week → 4 hours)
- ▶ Student Volunteer - FICTA'25 (Frontiers of Intelligent Computing: Theory and Applications) International Conference